LaB 2 Conversions Base 2, 8, 10, 16

1. $0.25_{10} \rightarrow$ Base 2, 8, 16

$0.25 \times 2 = 0.5$
$.5 \times 2 = 1.0$
$0.25_{10} = 0.01_2$

$0.25 \times 8 = 2.0$
$0.25_{10} = 0.2_8$

$0.25_{10} \times 16 = 4.0$
$0.25_{10} = 0.4_{16}$

$$\boxed{0.25_{10} = 0.01_2 = 0.2_8 = 0.4_{16}}$$

2. $0.25_8 \longrightarrow$ Base 2, 8, 16

$0.25_8 = (2 \times 8^{-1}) + (5 \times 8^{-2}) = \frac{2}{8} + \frac{5}{64} = \frac{21}{64} = 0.328125_{10}$

$0.25_8 = 0.328125_{10}$

$0.328125_{10} \times 2 = 0.65625$
$.65625 \times 2 = 1.3125$
$.3125 \times 2 = 0.625$
$.625 \times 2 = 1.25$
$.25 \times 2 = 0.5$
$.5 \times 2 = 1.0$
$0.328125_{10} = 0.010101_2$

$0.328125 \times 16 = 5.25$
$.25 \times 16 = 4$
$0.328125_{10} = 0.54_{16}$

$$\boxed{0.25_8 = 0.010101_2 = 0.328125_{10} = 0.54_{16}}$$

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3- $0.25_{16} \rightarrow$ Base 2,8,10

$0.25_{16} = (2 \times 16^{-1}) + (5 \times 16^{-2}) = 2/16 + 5/256 = \frac{37}{256} = 0.14453125_{10}$

$0.25_{16} = 0.14453125_{10}$

$0.14453125 \times 2 = 0.2890625$

$.2890625 \times 2 = 0.578125$

$.578125 \times 2 = 1.15625$

$.15625 \times 2 = 0.3125$

$.3125 \times 2 = 0.625$

$.625 \times 2 = 1.25$

$.25 \times 2 = 0.5$

$.5 \times 2 = 1.0$

$0.14453125_{10} = 0.00100101_2$

$0.14453125 \times 8 = 1.15625$

$.15625 \times 8 = 1.25$

$.25 \times 8 = 2.0$

$0.14453125_{10} = 0.112_8$

$$\boxed{0.25_{16} = 0.00100101_2 = 0.14453125_{10} = 0.112_8}$$

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4. $0.1101_2 \rightarrow$ Base 8, 10, 16

$$0.1101_2 = (1 \times 2^{-1}) + (1 \times 2^{-2}) + (1 \times 2^{-4}) = \frac{1}{2} + \frac{1}{4} + \frac{1}{16} = \frac{13}{16} = 0.8125_{10}$$

$0.1101_2 = 0.8125_{10}$

$0.8125 \times 8 = 6.5$

$.5 \times 8 = 4$

$0.8125_{10} = 0.64_8$

$0.8125_{10} \times 16 = 13$

$0.8125_{10} = D_{16}$

$$\boxed{0.1101_2 = 0.64_8 = 0.8125_{10} = D_{16}}$$

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